

SQL DATA STRUCTURE

1. DATABASE STRUCTURE

Component	Description
Database Software	Software used to manage databases. <i>Ex: MySQL, PostgreSQL</i>
Database	Collection of related data.
Schema	Collection of tables; acts as a database blueprint.
Table	Stores data in rows and columns.
Data	Actual information stored in tables.

2. TABLE

A table is the basic structure of a database. It manages **column details, indexes, relationships and data storage**.

3. DATA TYPES

Type	Examples
Number	INT, BIGINT, DECIMAL
Date/Time	DATE, DATETIME, TIMESTAMP
Text	CHAR, VARCHAR, TEXT
Special	BOOLEAN, BLOB, JSON

4. COLUMN ATTRIBUTES

Attribute	Purpose
NOT NULL	Prevents empty (NULL) values.
AUTO_INCREMENT	Generates numbers automatically.
DEFAULT	Provides a default value.

5. SQL BASICS

- **CREATE** – Creates database objects.
- **SHOW** – Displays database information.
- **ALTER** – Modifies table structure.
- **DROP** – Removes a table.
- **TRUNCATE** – Removes table data but keeps its structure.
- Every SQL statement ends with ;

DATABASE FLOW

Database Software → Database → Schema → Table → Data

Movie Database Management An Analytical Reporting Case Study:

#Create a database and use the database

```
create database Movies_db;
use Movies_db;

create table movies
(
  rel_year int,
  mov_name varchar(100),
  act_name varchar(50),
  lang varchar(30),
  dir_name varchar(100)
);
```

#Insert the data's

```
INSERT INTO movies VALUES
(1, 'Jai Bhim', 'Drama', 'Tamil', 2021, 164, 8.8),
(2, 'RRR', 'Action', 'Telugu', 2022, 187, 8.6),
(3, 'KGF Chapter 2', 'Action', 'Kannada', 2022, 168, 8.4),
(4, 'Vikram', 'Action', 'Tamil', 2022, 175, 8.3),
(5, 'Soorarai Pottru', 'Drama', 'Tamil', 2020, 153, 8.7),
(6, 'Drishyam 2', 'Thriller', 'Malayalam', 2021, 152, 8.2),
(7, 'Dangal', 'Sports', 'Hindi', 2016, 161, 8.4),
(8, 'Baahubali 2', 'Fantasy', 'Telugu', 2017, 171, 8.2),
(9, 'Andhadhun', 'Thriller', 'Hindi', 2018, 139, 8.2),
(10, 'Kantara', 'Drama', 'Kannada', 2022, 148, 8.3),
(11, 'Leo', 'Action', 'Tamil', 2023, 164, 7.8),
(12, 'Jawan', 'Action', 'Hindi', 2023, 169, 7.9),
(13, 'Pushpa', 'Action', 'Telugu', 2021, 179, 7.6),
(14, '3 Idiots', 'Comedy', 'Hindi', 2009, 170, 8.4),
(15, 'Premam', 'Romance', 'Malayalam', 2015, 156, 8.3);
```

1. Display all movies.

```
SELECT * FROM movies;
```

Result Grid		Filter Rows:		Edit:	Export/Import:	
movie_id	movie_name	genre	language	release_year	duration	rating
1	Jai Bhim	Drama	Tamil	2021	164	8.8
2	RRR	Action	Telugu	2022	187	8.6
3	KGF Chapter 2	Action	Kannada	2022	168	8.4
4	Vikram	Action	Tamil	2022	175	8.3
5	Soorai Pottru	Drama	Tamil	2020	153	8.7
6	Drishyam 2	Thriller	Malayalam	2021	152	8.2
7	Dangal	Sports	Hindi	2016	161	8.4
8	Baahubali 2	Fantasy	Telugu	2017	171	8.2
9	Andhadhun	Thriller	Hindi	2018	139	8.2
10	Kantara	Drama	Kannada	2022	148	8.3
11	Leo	Action	Tamil	2023	164	7.8
12	Jawan	Action	Hindi	2023	169	7.9
13	Pushpa	Action	Telugu	2021	179	7.6
14	3 Idiots	Comedy	Hindi	2009	170	8.4
15	Premam	Romance	Malayalam	2015	156	8.3
NULL	NULL	NULL	NULL	NULL	NULL	NULL

2. List movie name, genre, and rating.

```
SELECT movie_name, genre, rating FROM movies;
```

movie_name	genre	rating
Jai Bhim	Drama	8.8
RRR	Action	8.6
KGF Chapter 2	Action	8.4
Vikram	Action	8.3
Soorai Pottru	Drama	8.7
Drishyam 2	Thriller	8.2
Dangal	Sports	8.4
Baahubali 2	Fantasy	8.2
Andhadhun	Thriller	8.2
Kantara	Drama	8.3
Leo	Action	7.8
Jawan	Action	7.9
Pushpa	Action	7.6
3 Idiots	Comedy	8.4
Premam	Romance	8.3

3. Show distinct movie languages.

```
SELECT DISTINCT language FROM movies;
```

language
Tamil
Telugu
Kannada
Malayalam
Hindi

4. Retrieve unique actor names.

```
SELECT DISTINCT act_name  
FROM movies;
```

Solution:

act_name
Kamal Haasan
Mammootty
Rajinikanth
Mohanlal
Vijay
Silambarasan
Ajith Kumar
Suriya
Trisha
Nayanthara
Karthi
Dhanush
Dulquer Salm...
Amala Paul
Rashmika Ma...
Manju Warior
Sai Pallavi
Parvathy Thi...
Vijay Sethup...